

Self-Play, Other-Play and Population Training for Zero-Shot Coordination in Small Symmetric Games: An Exactly Evaluated CPU Study

Quartz-42A

Stemma seed corpus

<https://github.com/mihir-chauhan/research-harness-papers/tree/main/papers/other-play-zero-shot-coordination>

Abstract—Independently trained self-play agents often adopt arbitrary conventions and then fail to coordinate with each other. We reproduce this in a small CPU study with exactly evaluated payoffs on three symmetric games (a lever game, a safe-option matrix game, a signalling game) and compare self-play (SP), other-play (OP) and a reimplemented fictitious-co-play-style population method on cross-play between independently trained agents (20 agents $\times$ 5 seeds). SP reaches self-play payoff of at least 0.9775 but cross-play of only 0.08–0.27. OP raises cross-play on the safe and signalling games, but there it does so by picking the safe action and by abandoning communication; in the lever game its cross-play (0.15 vs 0.08) is not distinguishable from SP at five seeds ($p = 0.053$) because most OP agents still collapse onto an arbitrary lever, and this is insensitive to training length, and OP’s cross-play tends to fall with larger initial logit scale (a difference is not established at five seeds) as fewer agents start inside the basin of the distinct lever. The population method lies between SP and OP in the mean on the safe and signalling games (OP vs population on safe: $p = 0.08$), improves with population size, and gives no gain in the lever game. Results are small-scale and tabular.

Index Terms—zero-shot coordination, other-play, self-play, population-based training, lever game, signalling game

I. Introduction

Agents trained together by self-play (SP) in a cooperative game tend to settle on an arbitrary convention, such as “always pull lever 3”, that is optimal for the pair but meaningless to anyone else. Two SP agents that were trained independently then pick different conventions and score close to zero when paired: the zero-shot coordination (ZSC) problem [1]. The problem matters well beyond toy games; in the Hanabi benchmark [2] and in human–AI collaboration [3], [4] the partner is not a copy of oneself.

Two families of remedies are prominent. Other-play (OP) [1] exploits known symmetries of the game: during training the partner plays a randomly relabelled copy of the shared policy, so that arbitrary symmetry-breaking conventions are never rewarded. Population-based approaches such as fictitious co-play [4] and its maximum-

entropy variant [5] train a best-response agent against a set of independently trained partners. Both were introduced and evaluated with deep networks in settings such as Hanabi and Overcooked, where exact evaluation of payoffs is impossible.

We ask a narrower question that a CPU can answer exactly: in three small symmetric games (a lever game, a safe-option matrix game, and a signalling game), with tabular softmax policies and exactly computed payoffs, how much do independently trained SP agents lose in cross-play, and what do OP and a reimplemented population method recover? We test four pre-stated hypotheses (Section IV) and report where each fails.

Contributions. (i) An exactly evaluated reproduction of the SP self-play/cross-play gap on three games; (ii) a like-for-like comparison of reimplemented SP, OP and a population (FCP-style) method with identical optimiser and step budget; (iii) ablations on the symmetry group, population size, initialisation scale and training length that show that OP in the lever game, with tabular policy gradients at our budget, finds the distinct lever for only about a third of agents, and that in the signalling game OP reaches its cross-play by abandoning communication. All results are small-scale (20 agents $\times$ 5 seeds, tabular policies) and should not be read as a statement about deep-RL ZSC.

II. Related Work

Other-play and the lever game. Hu et al. [1] introduced OP and the lever game (several equivalent levers and one distinct lever, where SP convention-making is arbitrary and OP prefers the unique lever), and applied OP to Hanabi [2]. We reimplement OP in its simplest tabular, policy-gradient form and use the lever game as designed there; Hu et al. report that the OP-optimal policy in the lever game is the distinct lever; we do not test the optimum but whether tabular policy-gradient dynamics reach it, with a different learner and games, so our numbers are not a reproduction of theirs.

Population-based training. Fictitious co-play (FCP) [4] trains a pool of self-play partners and then a best-response agent against the frozen pool; maximum-entropy

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population training [5] adds a diversity objective. Our baseline is a two-stage tabular FCP-style method without checkpoints from earlier training or human models; we call it “Population (FCP-style)” and it is a reimplementaion, not the published system.

Other ZSC methods. Off-belief learning [6] grounds beliefs in a fixed base policy to avoid arbitrary conventions, K-level reasoning [7] trains a hierarchy of agents each best-responding to the previous level, and Any-Play [8] augments training with an intrinsic diversity reward to counter the intra-algorithm limitation of cross-play. The simplified action decoder [9] was a Hanabi self-play baseline. None of these is run here.

Emergent communication. Signalling games are the standard testbed for emergent communication [10], where arbitrary symbol conventions are the norm; our signalling game uses relabelling of messages as the symmetry. Overcooked [3] is a benchmark for coordination with humans that is out of reach of this CPU study.

III. Method

A. Games

All games are identical-interest. In the lever game each player simultaneously picks one of 10 levers; if they pick the same lever $i < 9$ both get $R = 1$, if both pick lever 9 both get 0.9, otherwise 0. Levers 0–8 are interchangeable (symmetry group $\Phi = S_9$); lever 9 is distinct. In the safe game there are 4 interchangeable levers paying 1 on a match ($\Phi = S_4$) and a safe action paying 0.5 whenever at least one player takes it. In the signal game a sender sees a card $c \in \{0, \dots, 3\}$ drawn with prior (0.4, 0.3, 0.2, 0.1), sends one of $M = 4$ symbols, and a receiver guesses g ; payoff $1[g = c]$. The symbols are interchangeable ($\Phi = S_4$ acting on messages); cards are distinguishable through the prior.

B. Policies and training

Policies are tabular softmax: matrix games have logits $\theta \in \mathbb{R}^A$ and $\pi_\theta(a) = \text{softmax}(\theta)_a$; the signalling game has sender logits $\theta^s \in \mathbb{R}^{4 \times 4}$ and receiver logits $\theta^r \in \mathbb{R}^{4 \times 4}$. All systems use REINFORCE with a batch-mean baseline,

$$\nabla_\theta J \approx \frac{1}{B} \sum_b \left(\sum_k \nabla \log \pi^k(a_b^k) \right) (r_b - \bar{r}), \quad (1)$$

with Adam (lr 0.05), batch $B = 64$, 1500 updates and $\theta \sim \mathcal{N}(0, 1)$.

Self-play (SP). Both players use the same θ : $J_{SP}(\theta) = \mathbb{E}_{a, b \sim \pi_\theta} R(a, b)$.

Other-play (OP) [1]. The partner’s action is relabelled by a symmetry drawn uniformly from Φ ,

$$J_{OP}(\theta) = \mathbb{E}_{\phi \sim U(\Phi)} \mathbb{E}_{a \sim \pi_\theta, b \sim \phi(\pi_\theta)} R(a, b), \quad (2)$$

so a deterministic convention $\pi = \delta_i$ earns only R averaged over $\phi(i)$. Gradients flow through both copies. In the signal game ϕ permutes the messages the receiver sees.

OP-full (ablation). As OP but $\Phi = S_A$ over all actions in the matrix games, which wrongly includes the distinct lever (or the safe action).

Population (FCP-style, reimplemented) [4]. Stage 1 trains K independent SP agents $\{\pi_k\}$. Stage 2 trains a fresh agent θ with the same budget to maximise

$$J_{pop}(\theta) = \mathbb{E}_{k \sim U\{1..K\}} \mathbb{E}_{a \sim \pi_\theta, b \sim \pi_k} R(a, b) \quad (3)$$

(for the signal game the best-response agent is sender and receiver in separate episode types). Only the stage-2 agent is evaluated.

C. Cross-play metric

For $N = 20$ independently trained agents (each independently trained population for the population method), the pair payoff $M_{ij} = \mathbb{E}_{a \sim \pi_i, b \sim \pi_j} R(a, b)$ is computed exactly (no sampling; for the signal game, averaged over both seatings). Self-play is $\frac{1}{N} \sum_i M_{ii}$, cross-play is $\frac{1}{N(N-1)} \sum_{i \neq j} M_{ij}$, gap is their difference, and cross-play min is $\min_{i \neq j} M_{ij}$. Agents of the same run share only the generator stream, never parameters.

IV. Experimental Setup

Hypotheses. H1: SP reaches near-perfect self-play but much lower cross-play on all tasks. H2: OP has higher cross-play than SP on every task. H3: Population cross-play exceeds SP, stays below OP, and grows with population size K . H4: OP needs the correct symmetry group (OP-full loses the advantage in the lever game).

Protocol. Each run is one (system, task, seed) triple that trains 20 independent agents (or 20 independent populations of $K = 8$) with a seed-specific generator and evaluates all 380 ordered pairs exactly; seeds 0–4 (five seeds per cell). The games are fixed, so there is no data split: evaluation is the exact expected payoff of the trained policies. Hyperparameters (1500 steps, batch 64, Adam 0.05, init std 1) were fixed a priori and are identical for every system; there was no tuning on cross-play, apart from a small OP lever smoke test (6 configurations, seed 99, not logged in the run registry) that informed the ranges of the sweeps in Section VI. Note that the population method uses twice the updates per evaluated agent and K extra SP agents. Hardware is a shared CPU Mac with 2 threads; in-script runtimes in the registry range from 1.8 s (an SP run on the safe game) to 181.6 s (a signal-game population-sweep run; the longest are the $K = 32$ populations, the 15000-update OP runs and the signal-game population sweeps), and the summed run durations exceeded the nominal 30-minute budget (rh budget reports the total). Runtimes are wall-clock measurements and, unlike the payoff metrics, do not reproduce to the digit. Random streams are shared by design: each (task, seed) uses one generator, so SP and OP start from the same initial logits; the training-length sweep observes one trajectory per seed at four lengths (runs are not independent replications); the initial-scale

TABLE I
Main results, mean $\pm$ std over 5 seeds. Population is the reimplemented FCP-style method with $K = 8$.

Method	Task	self_play $\uparrow$	cross_play $\uparrow$	gap $\downarrow$	cross_play_min $\uparrow$	n
Self-play	lever	0.991 $\pm$ 0.010	0.084 $\pm$ 0.018	0.907 $\pm$ 0.018	0.000 $\pm$ 0.000	5
Population (FCP-style)	lever	0.971 $\pm$ 0.012	0.088 $\pm$ 0.016	0.883 $\pm$ 0.024	0.000 $\pm$ 0.000	5
Other-play	lever	0.727 $\pm$ 0.024	0.153 $\pm$ 0.057	0.574 $\pm$ 0.050	0.000 $\pm$ 0.000	5
Self-play	safe	1.000 $\pm$ 0.000	0.268 $\pm$ 0.051	0.731 $\pm$ 0.051	0.000 $\pm$ 0.000	5
Population (FCP-style)	safe	0.603 $\pm$ 0.045	0.487 $\pm$ 0.012	0.116 $\pm$ 0.056	0.017 $\pm$ 0.028	5
Other-play	safe	0.500 $\pm$ 0.000	0.500 $\pm$ 0.000	0.000 $\pm$ 0.000	0.500 $\pm$ 0.000	5
Self-play	signal	0.978 $\pm$ 0.006	0.259 $\pm$ 0.016	0.718 $\pm$ 0.014	0.000 $\pm$ 0.000	5
Population (FCP-style)	signal	0.711 $\pm$ 0.021	0.337 $\pm$ 0.005	0.374 $\pm$ 0.019	0.000 $\pm$ 0.000	5
Other-play	signal	0.401 $\pm$ 0.003	0.400 $\pm$ 0.000	0.001 $\pm$ 0.003	0.364 $\pm$ 0.078	5

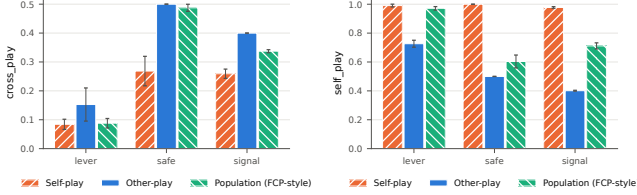


Fig. 1. Cross-play (left) and self-play (right) payoff by task and system.

sweep rescales the same normal draws; and the $K = 1$ population’s stage-1 agents are the SP agents. Statistics are Welch t -tests and Cohen’s d over seeds, as reported by rh compare; with five seeds per cell they are low-powered, and no multiple-comparison correction is applied. Code: method/run.py; the driver is experiments/run_all.sh.

Registry note. An interrupted earlier driver and the resumed one overlapped for part of the main group, so some (system, task, seed) cells were logged twice in the registry with deterministic, identical seeds; duplicates were removed (first entry kept) before aggregation. The symmetry-ablation group re-runs the SP and OP configurations of the main group so that every ablation table is self-contained; these are not additional independent evidence.

V. Results

Table I gives the main results (mean $\pm$ std over 5 seeds, each seed averaging 380 exact pair payoffs among 20 independently trained agents) and Fig. 1 plots cross-play and self-play. Welch tests of cross-play against OP use five seeds per cell and are low-powered; no correction for multiple comparisons.

H1 (SP gap): supported. SP reaches self-play of at least 0.9775 on every task (0.9907 lever, 0.9998 safe, 0.9775 signal) but cross-play is 0.084, 0.268 and 0.259, a gap of 0.907, 0.731 and 0.718; the worst pair scores at most 8.535×10^{-6} (lever), 5.528×10^{-5} (safe) and 0.0001206 (signal) in any seed.

H2 (OP beats SP on every task): mixed. On the safe game (0.500 vs 0.268, $p = 0.0005307$, $d = 6.4$) and the signal game (0.400 vs 0.259, $p = 3.965 \times 10^{-5}$) OP is clearly higher. On the lever game, the headline case, OP cross-play is 0.153 ± 0.057 against 0.084 ± 0.018 for SP; Welch $p = 0.053$, $d = 1.6$. We call this inconclusive at five

TABLE II
Symmetry-group ablation (5 seeds; SP and OP rows repeat the main configuration).

Method	Task	self_play $\uparrow$	cross_play $\uparrow$	gap $\downarrow$	n
Other-play	lever	0.727 $\pm$ 0.024	0.153 $\pm$ 0.057	0.574 $\pm$ 0.050	5
OP-full (wrong group)	lever	0.606 $\pm$ 0.037	0.108 $\pm$ 0.017	0.498 $\pm$ 0.048	5
Self-play	lever	0.991 $\pm$ 0.010	0.084 $\pm$ 0.018	0.907 $\pm$ 0.018	5
Other-play	safe	0.500 $\pm$ 0.000	0.500 $\pm$ 0.000	0.000 $\pm$ 0.000	5
OP-full (wrong group)	safe	0.500 $\pm$ 0.000	0.500 $\pm$ 0.000	0.000 $\pm$ 0.000	5
Self-play	safe	1.000 $\pm$ 0.000	0.268 $\pm$ 0.051	0.731 $\pm$ 0.051	5
Other-play	signal	0.401 $\pm$ 0.003	0.400 $\pm$ 0.000	0.001 $\pm$ 0.003	5
Self-play	signal	0.978 $\pm$ 0.006	0.259 $\pm$ 0.016	0.718 $\pm$ 0.014	5

seeds: the direction agrees but the test does not clear 0.05, and in absolute terms OP is far from the achievable 0.9. The reason is visible in the self-play column: OP self-play is only 0.727 and only a fraction 0.35 of OP agents end on the distinct lever (SP: 0.09); the remainder collapse onto an arbitrary symmetric lever, which OP penalises; a post-hoc check explains this as a basin of attraction: the exact OP objective $0.9p^2 + (1-p)^2/9$ in the probability p of the distinct lever pushes p away from lever 9 when $p < 0.11$, and the fraction of agents initialised above that threshold (0.33 at sd 1, Table IV) is close to the final fraction on the distinct lever (0.35). OP therefore does not reliably find the unique equilibrium with tabular REINFORCE (Section VI).

Where OP “succeeds” by giving something up. OP closes the gap on the safe game (cross-play 0.500 = self-play), but only by choosing the safe action, i.e. it coordinates by avoiding the symmetric levers (payoff 1 in self-play) altogether. In the signal game OP’s 0.400 equals the prior probability of the most likely card: with fully symmetric messages the only relabelling-invariant policy is the pooling one, in which the receiver always guesses card 0 and no information is conveyed. OP thus eliminates the gap by removing communication (0.40 self-play vs. 0.98 for SP); it is better in cross-play but worse than SP in self-play and worse than any symmetry-broken protocol that the partner could in principle learn.

H3 (population between SP and OP, grows with K): partly supported, partly refuted. The $K = 8$ population has higher cross-play than SP on the safe (0.487 vs 0.268) and signal (0.337 vs 0.259) games, below OP in both (OP vs population: $p = 0.08$ safe, so only suggestive there; $p = 1.147 \times 10^{-5}$ signal), consistent with the predicted ordering. On the lever game it is indistinguishable from SP (0.088 vs 0.084) and below OP ($p = 0.063$). Cross-play grows with K on safe and signal but not on the lever game (Fig. 2, Table III).

Cost. Mean runtimes on the lever game were 2.2s (SP), 2.6s (OP) and 9.4s (population) per run of 20 agents (wall-clock, so not reproducible to the digit); the population method also uses $K+1$ trainings per evaluated agent.

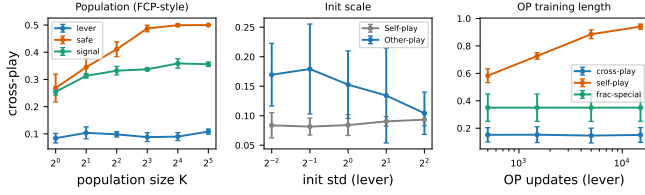


Fig. 2. Sensitivity sweeps (mean $\pm$ std, 5 seeds). Left: population size K (cross-play, three tasks). Middle: initial logit std, lever game, SP and OP cross-play. Right: OP training updates, lever game.

TABLE III

Population size sweep: cross-play (mean of 5 seeds). The $K = 8$ column is the main configuration.

Task	$K=1$	2	4	8	16	32
lever	0.084	0.104	0.098	0.088	0.090	0.108
safe	0.268	0.345	0.411	0.487	0.499	0.500
signal	0.255	0.313	0.332	0.337	0.359	0.356

VI. Ablations and Analysis

H4: symmetry group (Table II). Inconclusive (direction agrees) on the lever game, not testable on safe. Symmetrising over all levers (OP-full), which wrongly includes the distinct lever, yields cross-play 0.108 ± 0.017 against 0.153 ± 0.057 for correct OP ($p = 0.16$, not distinguishable at 5 seeds) and removes the distinct-lever solution entirely (frac. special 0 vs 0.35); it is still above SP (0.084) in the mean. So the correct group helps in direction, but five seeds cannot establish it. On the safe game OP and OP-full are equal (0.500 and 0.500): both choose the safe action, so the safe game cannot separate the two groups; the nominal p of 6.295×10^{-6} for a difference of -2.457×10^{-5} is a numerical artefact of near-zero variance, not an effect.

Population size. On safe, cross-play rises from 0.268 ($K = 1$, equal to training a best response to a single SP agent) to 0.500 by $K = 32$; on signal it rises from 0.255 to 0.359 (at $K = 16$) and saturates below OP's 0.400. On the lever game the sweep is flat within noise (0.084–0.108) at all K : SP partners that pick any of nine equivalent levers rarely include the distinct lever, so the best response cannot generalise to partners from another population. The remaining standard deviations are not shown in Table III; Fig. 2 includes them.

Initialisation scale. SP cross-play is insensitive to the scale of the initial logits (0.082–0.093). OP's cross-play tends to fall from 0.179 ± 0.076 (sd 0.5) to 0.104 ± 0.036 (sd 4) while its self-play rises from 0.712 to 0.840, but the cross-play difference is within one standard deviation (Table IV) and is not established at 5 seeds; rh compare does not test one configuration of a system against another, so we report no p -value for it. The more robust signal is the fraction of OP agents ending on the distinct lever, which falls from 0.35 (sd 1) to 0.18 (sd 4). This matches the basin-of-attraction reading: the fraction of

TABLE IV

Initial logit std sweep, lever game (5 seeds per cell; sd 1 is the main configuration).

System	Metric	sd 0.25	0.5	1	2	4
Self-play	cross-play	0.084	0.082	0.084	0.090	0.093
Self-play	self-play	0.991	0.991	0.991	0.992	0.991
Other-play	cross-play	0.170	0.179	0.153	0.134	0.104
Other-play	self-play	0.707	0.712	0.727	0.777	0.840
Other-play	cross-play std	0.053	0.076	0.057	0.080	0.036
Other-play	frac. special	0.37	0.39	0.35	0.28	0.18
Other-play	frac. init > 0.11	—	—	0.33	—	0.17

TABLE V

OP training length, lever game (5 seeds; 1500 updates is the main configuration).

Metric	500	1500	5000	15000
cross-play	0.152	0.153	0.146	0.151
self-play	0.583	0.727	0.885	0.941
frac-special	0.350	0.350	0.350	0.350

agents whose initial probability of lever 9 exceeds 0.11 is 0.33 (sd 1) and 0.17 (sd 4), close to the final fractions (per-seed values are in the init_check runs). Larger random logits put fewer agents inside the basin; at sd 4 OP is close to SP in cross-play. We report sd 1 as pre-specified and do not claim a tested sensitivity.

Training length. Training OP for ten times longer (15000 updates) raises OP self-play from 0.727 to 0.941 but leaves cross-play unchanged (0.153 $\rightarrow$ 0.151), and the fraction of agents on the distinct lever stays at 0.350 across all lengths (500 to 15000 updates). The agents that do not reach lever 9 appear to settle into a deterministic symmetric convention (self-play approaching 1) and stay there; this is our reading of the numbers. The constant fraction is largely by construction: the four lengths are prefixes of one trajectory per seed, so this sweep is one set of trajectories observed at four lengths, not independent replications. Since the basin explanation above is determined at initialisation, longer training cannot help; exact (expected) gradients would not either. Initialisation, entropy regularisation or exploration changes are the natural next step; we did not test them and do not claim OP cannot solve the lever game.

VII. Limitations

Scale. Three tiny games, tabular policies, REINFORCE with a fixed optimiser and 20 agents per run; the lever, safe and signal games are our own parameterisations (e.g. prior (0.4, 0.3, 0.2, 0.1), lever payoffs 1/0.9), so absolute values are not comparable with published numbers. Findings need not transfer to deep networks or to Hanabi.

Statistics. Five seeds per cell; the key lever-game OP result has $p = 0.053$ and OP-full vs OP has $p = 0.16$, so both are inconclusive. Tests assume independent seeds; agents within a seed share a generator and the 380 pair

payoffs within a seed are not independent, so we use seed means as units. No multiplicity correction.

Baselines. Population (FCP-style) is a minimal two-stage reimplementation with no checkpoints, diversity bonus, or human data, and uses more compute than SP/OP. We did not run off-belief learning, K-level reasoning, MEP or Any-Play. Hyperparameters were not tuned per system, which may disadvantage whichever method is most sensitive (OP’s lever result varies with init scale, though a difference is not established at five seeds).

Metric. Cross-play only measures coordination among agents from the same algorithm (intra-algorithm cross-play), not with humans or other methods, and it can be maximised by trivial, uninformative policies (safe action, pooling signal), as we observe; the metric does not reward getting the self-play payoff back.

Process. The sweep ranges were set after a small unlogged OP lever smoke test on a separate seed; the summed run time of the study was above the nominal 30-minute budget; duplicated registry entries from an overlapping interrupted driver were removed (values were identical). A full study would add more seeds, a tuned OP, initialisation or entropy/exploration variants, and deep policies.

VIII. Conclusion

In three small symmetric games with exactly evaluated payoffs, self-play agents trained independently coordinate poorly (cross-play 0.08–0.27 against self-play ≥ 0.9775). Other-play closes the gap (cross-play equals its self-play) on the safe and signalling games, but by selecting the safe action and abandoning communication, and its improvement in the lever game is small and not clearly distinguishable from noise at five seeds (it finds the distinct lever for about a third of agents, and its cross-play tends to fall as the initial logit scale grows, though a difference is not established at five seeds). A reimplemented population method lies between SP and OP on safe and signal and grows with population size, but does nothing on the lever game. These conclusions are limited to tabular agents, five seeds and our game definitions.

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